

COMPUTER TECHNOLOGY

Computer technology is an expanding branch of engineering, with roots in both electrical engineering and computer science. It includes the related areas of computer hardware and software; computer interfacing, programming, and networking; analog and digital electronics; and robotics. Computer technicians, technologists, and engineers work in every sector of society, in careers ranging from building and repairing computer systems to designing and installing computer networks, to designing and building prototype robots and electronic devices.

Students taking computer technology courses will learn to work safely with computer equipment, electronic circuits, and robotic devices, and will explore the Essential Skills and work habits that are important for success in computer technology. Computer technology courses prepare students for apprenticeship, further study at college or university, or entry into the workplace directly after graduation.

The list of approved emphasis areas for computer technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Computer Technology, Grade 12

Workplace Preparation

TEJ4E

This course enables students to further develop their practical understanding of computer hardware, software, networks, and operating systems. Students will use utility and application software, and will follow proper procedures for installing, maintaining, and troubleshooting computer systems and networks. In addition to demonstrating an understanding of the ethical use and environmental effects of computers, students will develop marketable skills and assess career opportunities in the field.

Prerequisite: Computer Technology, Grade 11, Workplace Preparation

A. COMPUTER TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe the function and development of a variety of current computer hardware;
- A2.** describe network topologies, devices, and connection media as well as common user network requirements;
- A3.** describe various types of software, analyse software needs, and evaluate available software.

SPECIFIC EXPECTATIONS

A1. Computer Hardware

By the end of this course, students will:

- A1.1** accurately use relevant technical terminology to describe the specifications of computer hardware (*e.g., type, speed, capacity, compatibility, connections*);
- A1.2** describe the characteristics of processor types (*e.g., 32-bit, 64-bit, multi-core*);
- A1.3** describe the operation of current input and output devices (*e.g., keyboards, mice, tablets, printers, monitors, scanners, webcams*);
- A1.4** describe the evolution of home computing (*e.g., cost, availability, and ease of use of computing systems*);
- A1.5** describe hardware innovations in computers and related technologies (*e.g., lower costs, faster speeds, smaller sizes, and greater memory density for computers and for cellular, hand-held, and biometrics devices*);
- A1.6** explain the effect of trends in software design and data storage on hardware requirements and data processing (*e.g., the need for increased processor speed, memory, storage capacity, and bandwidth; longer boot times; large increases in the number and size of files*).

A2. Networking Concepts

By the end of this course, students will:

- A2.1** research and describe common network topologies and technologies;

- A2.2** compare hardware and connection media (*e.g., hardware: hub, switch, router; media: UTP, fibre-optic cable, wireless*) used for different types of networks (*e.g., home and small office, small-to-medium enterprise*);

- A2.3** describe common user requirements that affect the design of a network (*e.g., shared printer, wireless access, shared Internet connection, remote access*).

A3. Software

By the end of this course, students will:

- A3.1** describe the purpose and basic operation of an operating system;
- A3.2** describe the purpose and basic operation of common application software (*e.g., word processors, spreadsheets, databases, programming environments*);
- A3.3** describe the purpose and basic operation of various types of utility software (*e.g., system tools, backup and recovery software, antivirus and anti-spyware programs, security suites*);
- A3.4** analyse user software needs for a given purpose;
- A3.5** evaluate the suitability of available software for a specific task;
- A3.6** describe the specific minimum hardware configurations required for various software.

B. COMPUTER TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** determine and report on hardware solutions for user computing needs;
- B2.** install, maintain, and troubleshoot computer hardware, and design backup procedures;
- B3.** install, configure, manage, maintain, and troubleshoot computer networks and related services;
- B4.** install, configure, and update a variety of software.

SPECIFIC EXPECTATIONS

B1. Hardware Solutions

By the end of this course, students will:

- B1.1** use a problem-solving process (see pp. 21–23) to find solutions for user hardware needs, and compare and contrast solutions for various situations (*e.g., home computing, desktop publishing, small business, large office*);
- B1.2** compare hardware availability and costs from local, national, and global suppliers;
- B1.3** write a report recommending computing hardware to meet user requirements.

B2. Installation, Maintenance, and Troubleshooting

By the end of this course, students will:

- B2.1** develop and follow procedures for hardware installation, service, and troubleshooting;
- B2.2** document and follow correct procedures to prevent damage to computer components (*e.g., use of anti-static wrist straps, mats, bags, and containers*);
- B2.3** perform preventive maintenance on a variety of hardware components;
- B2.4** use utility software and/or diagnostic tools to correct problems on a computer and/or a network;
- B2.5** design effective procedures for backing up system data and user information.

B3. Network Setup and Management

By the end of this course, students will:

- B3.1** develop and follow procedures for network installation, service, and troubleshooting;
- B3.2** set up and/or configure networked workstations and shared devices using appropriate connection media (*e.g., UTP straight-through and cross-over cables, serial cables, fibre optics, wireless*);
- B3.3** install and configure network operating systems and client services;
- B3.4** use network utility software (*e.g., protocol analyser, extended ping, extended traceroute*) to diagnose and correct problems.

B4. Software Implementation

By the end of this course, students will:

- B4.1** install and configure new software and upgrades on a computer system;
- B4.2** describe the need for software, firmware, and device-driver upgrades, identify various sources for them, and follow proper procedures for installing the various types of upgrades;
- B4.3** compare software availability and costs from local, national, and global suppliers;
- B4.4** use manuals and online documentation to explore the features of new software.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** describe environmental issues related to the widespread use of computer technology, and apply strategies to reduce environmental harm from computer use;
- C2.** analyse societal issues related to the widespread use of computer technology.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** assess the effects of computer technology on the environment (*e.g., leakage of hazardous substances from obsolete computers dumped in landfills or improperly recycled; increased energy use; benefits of computer-controlled heating and cooling systems*);
- C1.2** outline and apply strategies to recycle and reuse computer components (*e.g., build computers using used components and donate to a community partner, offer a service where computers can be upgraded using used components*);
- C1.3** describe and apply strategies and devices that help reduce the energy used by computers at home and in the workplace (*e.g., software that throttles drive speed and CPU speed, monitors that turn off automatically, more efficient processors, lower-speed hard drives, diskless computers, virtualization to eliminate extra computers*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** analyse the benefits of computer technology for society (*e.g., improved access to technology for economically disadvantaged people and nations, greater efficiency and lower costs for information services, development of a “global village”, use of computers to help monitor and predict long-term environmental changes*);
- C2.2** analyse the drawbacks of computer technology for society (*e.g., Internet gambling addictions, more sedentary lifestyle, spam, telemarketing, loss of privacy*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** explain and follow computer-related safety standards and practices;
- D2.** describe ethical and security issues related to the use of computers;
- D3.** demonstrate an understanding of professional customer-service practices;
- D4.** apply the skills required for success in the workplace;
- D5.** describe opportunities for careers and training related to computer technology, and explain the need for lifelong learning in the computer technology industry.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** explain the importance of following industry health and safety standards and practices (*e.g., standards and regulations specified in the Workplace Hazardous Materials Information System [WHMIS], the Electrical Safety Code, and the Occupational Health and Safety Act, and by the Workplace Safety and Insurance Board [WSIB]; ergonomically sound workplace arrangements and practices*);
- D1.2** describe and use appropriate equipment, techniques, and strategies to avoid health and safety problems when assembling, using, and maintaining computer systems (*e.g., repetitive strain injuries, eye strain, electrical shock*);
- D1.3** research and discuss issues related to Internet safety (*e.g., protection of information stored on computers or transmitted over a network, cyberstalking, cyberbullying, privacy policies*).

D2. Ethics and Security

By the end of this course, students will:

- D2.1** describe the components of an acceptable-use policy for computers (*e.g., restrictions on commercial or personal use, prohibition of inappropriate content and plagiarism, protection of privacy and intellectual property rights*);
- D2.2** explain the reasons for software licensing agreements and registration procedures;

- D2.3** explain the importance of computer security (*e.g., passwords, security software updates, protecting personal identity information and client data*).

D3. Customer Service

By the end of this course, students will:

- D3.1** develop procedures for tracking client data electronically (*e.g., using a spreadsheet, database, journal, or log*);
- D3.2** explain the importance of professionalism in customer relations (*e.g., ensuring appropriate personal appearance, using active listening techniques, making eye contact, speaking clearly and respectfully, being approachable, being aware and respectful of diverse cultural communication styles*);
- D3.3** develop and model customer-service procedures for dealing with clients (*e.g., procedures for complaints, troubleshooting, and providing customer support by telephone, email, or the Internet*);
- D3.4** communicate with clients using an appropriate level of technical terminology;
- D3.5** model user-level support for software (*e.g., simulate an IT help desk, create an FAQ website*).

D4. Workplace Skills

By the end of this course, students will:

- D4.1** demonstrate time-management skills in project settings (*e.g., set realistic goals, recognize time constraints, plan for deadlines, prioritize tasks*);

D4.2 conduct and participate in all aspects of effective meetings for various purposes (*e.g., create and follow an agenda, write and circulate minutes, conduct chaired and round-table meetings*);

D4.3 use computer terminology correctly, and compile a glossary of computer terms and acronyms.

D5. Career Opportunities

By the end of this course, students will:

D5.1 explore various computer-related job opportunities in local, national, and international businesses and industries (*e.g., retail salesperson, IT hardware technician, IT network technician, electronic service technician*);

D5.2 describe the opportunities for and the importance of postsecondary training and certification related to computer technology (*e.g., apprenticeship, college courses, trade certifications*);

D5.3 explain the need for lifelong learning in the computer technology industry;

D5.4 demonstrate an understanding of and apply the Essential Skills that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (OSP) (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);

D5.5 demonstrate an understanding of and apply the work habits that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability, organization, working independently, initiative, self-advocacy, customer service*);

D5.6 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in computer technology (*e.g., Passport to Safety certificate, OSP Work Plan, OSP Transition Plan, work logs, photographs of projects*), and explain why having a current portfolio is important for career development and advancement.